Triglycerides

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

RELATED INFORMATION

- Cardiovascular Disease Risk Assessment Tools (/node/462)

Use

Evaluate turbid samples of blood, plasma, and serum; work up of chylomicronemia; evaluate hyperlipidemia; occasional cases of diabetes mellitus and/or pancreatitis are detected by hypertriglyceridemia. High levels may occur with hypothyroidism, nephrotic syndromes, carbohydrate-sensitive hypertriglyceridemia, glycogen storage disease, and in hyperlipoproteinemias type I, IIb, III, IV, and V. Some alcoholics have hypertriglyceridemia which disappears with abstinence. Extremely high triglyceride levels may occur with alcohol abuse. Triglyceride is needed for calculation of LDL-C (low-density lipoprotein cholesterol) concentration. Disturbances in triglyceride metabolism relate to diabetes and are a risk factor for atherosclerotic disease, but not an independent one.[1]

Although the role of hypertriglyceridemia as a risk factor for coronary arterial disease has been controversial, a more consistent association for women exists and analysis of preliminary data supports triglyceride levels as a predictor in men with lower LDL cholesterol levels[2] and with cholesterol values <220 mg/dL.[3]

In familial combined hyperlipidemia, hypertriglyceridemia may be found before hypercholesterolemia. Nevertheless, a strong case is not available for primary triglyceride evaluation of healthy persons without positive family history of coronary disease or other risk factors. Some knowledgeable authorities favor testing with lipid panels, including triglycerides, for reasons discussed elsewhere in this listing.

In exogenous hypertriglyceridemia, chylomicrons float as a layer in the tube of refrigerated, stored serum.

Limitations

If triglyceride is >800 mg/dL, LDL cannot be calculated accurately by the NIH formula. Some women on estrogens and high estrogen oral contraceptives have an increase of triglyceride. Increases occur with pregnancy, similar to those with oral contraceptives. The most common cause of triglyceride increase is inadequate patient fasting. Hypertriglyceridemia is associated with use of thiazide diuretics and β-adrenergic blocking agents.

Custom Additional Information

Triglycerides commonly increase with obesity and may increase with chronic renal or liver disease. A positive association exists between diabetes mellitus and hypertriglyceridemia. Extremely high triglyceride levels suggest the possibility of pancreatitis. **Chylomicronemia**, although associated with pancreatitis, is not accompanied by increased atherogenesis. Chylomicrons are not seen in normal fasting serum, but are found in the sera of normal subjects following a fatty meal as exogenous triglycerides. Left refrigerated, chylomicrons float to the surface of a sample overnight; VLDL remain in suspension. Triglyceride physiologically is carried mostly as very low-density lipoproteins (VLDL). The triglyceride in VLDL is endogenous from hepatic synthesis.

When turbidity of blood, serum, or plasma is seen, triglyceride is often >350 mg/dL. **Fasting chylomicronemia** occurs with but is not limited to deficiency of apo-CII (apolipoprotein work-up). It occurs also with deficiency of **lipoprotein lipase**, an enzyme.

A positive association exists between gout and hypertriglyceridemia.

Drug effects have been summarized.[4]

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

• Normal: <150 mg/dL

• Borderline-high: 150−199 mg/dL

• High: 200−499 mg/dL

• Very high: >500 mg/dL

Storage Instructions

Maintain specimen at room temperature.

Patient Preparation

Patient should fast overnight (12 hours preferred; eight hours acceptable) preceding collection of specimen.

Causes for Rejection

Specimen collected in a glycerinated tube; improper labeling

Footnotes

1. Rifkind BM, Segal P. Lipid Research Clinics Program reference values for hyperlipidemia and hypolipidemia. *JAMA.* 1983 Oct 14; 250(14):1869-1872. PubMed 6578354 (http://www.ncbi.nlm.nih.gov/pubmed/6578354)

2. Brunzell JD, Austin MA. Plasma triglyceride levels and coronary disease. *N Engl J Med.* 1989 May 11; 320(19):1273-1275. PubMed 2710207 (http://www.ncbi.nlm.nih.gov/pubmed/2710207)

3. Faulkner WR. Triglycerides—What do they mean? *Lab Report for Physicians.* 1987; 9:49-52.

4. Steinmetz J, Jouanel P, Thuillier Y. Triglycerides. In Siest G, Galteau MM, eds. *Drug Effects on Laboratory Test Results Analytical Interferences and Pharmacological Effects.* Littleton, Mass: PSG Publishing Co Inc; 1988:405-422.